Mouhammed Soumaille

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Mathematics and computer engineering undergraduate with research experience in probabilistic modeling and empirical evaluation in Python. Interested in empirical AI safety, especially rigorous evaluation of model behavior and interpretability, where probability and statistics can make results more trustworthy.

Education

American University of Beirut (AUB) Expected Spring 2029
B.S. in Computer Science and Engineering & Pure Mathematics *Beirut, Lebanon*

- **Relevant coursework:** Machine Learning; Graduate Probability Theory; Graduate Convex Optimization; Theoretical Foundations of Security & Privacy; Complexity Theory; Data Structures & Algorithms; Real Analysis; Advanced Linear Algebra

Research Experience

Research Assistant — Heavy-Tailed Financial Risk Fall 2025 – Spring 2026
American University of Beirut *Advisors: Prof. Ibrahim Abou Faycal, Prof. Jihad Fahs*

- Developed the **Extreme Loss Quantile Bound (ELQB)**, a spectral risk measure for extreme losses under heavy-tailed distributions; proved coherence, normalization, finiteness, and limiting behavior.
- Extended the framework to portfolio-level and conditional settings, introducing **Co-ELQB** and a systemic amplification ratio that quantifies how much an asset's tail risk grows under system-wide stress.
- Built the full empirical pipeline in **Python** (empirical quantile estimation, numerical integration) and evaluated it on five large-cap U.S. equities with daily data from 2020–2025.
- Ran a rolling-window analysis of the COVID-19 market crash to track how the measures respond to system-wide stress; documented results in a research report and slides.

Research Assistant — Quantum Error Correction Fall 2026 – Present
American University of Beirut *Advisor: Prof. Wael Itani*

- Computing entanglement polytopes of multipartite quantum systems from reduced density matrices and local spectra, using group-theoretic techniques; studying their relevance to quantum error-correcting codes.

Machine Learning Projects

DAYFIT — Visual Clothing Search Fall 2026
American University of Beirut *In progress*

- Building an image-retrieval system that returns the top-5 most similar garments from a 1,000–3,000 image catalog using pretrained **CLIP** embeddings.
- Designing the evaluation: comparing CLIP against a color-histogram baseline on Precision@5 and latency, and stress-testing **robustness** to shifts in lighting, background, and camera angle.
- Serving the model through a FastAPI endpoint with a Streamlit interface, packaged with Docker.

Technical Skills

Programming: Python (primary), C++, MATLAB
ML & scientific computing: PyTorch, scikit-learn, NumPy, SciPy, pandas, CVXPY
Tools: Git, Linux/Bash, Docker, FastAPI, Streamlit, LaTeX

Teaching & Communication

Tutor — Probability & Statistics; Analysis I Fall 2026
Mathematics Tutoring Center, AUB

- Lead review sessions explaining probabilistic reasoning and proof-based analysis clearly to students at different levels.